

Drone Detection, Tracking and Trajectory Prediction

System Architecture and Flowcharts

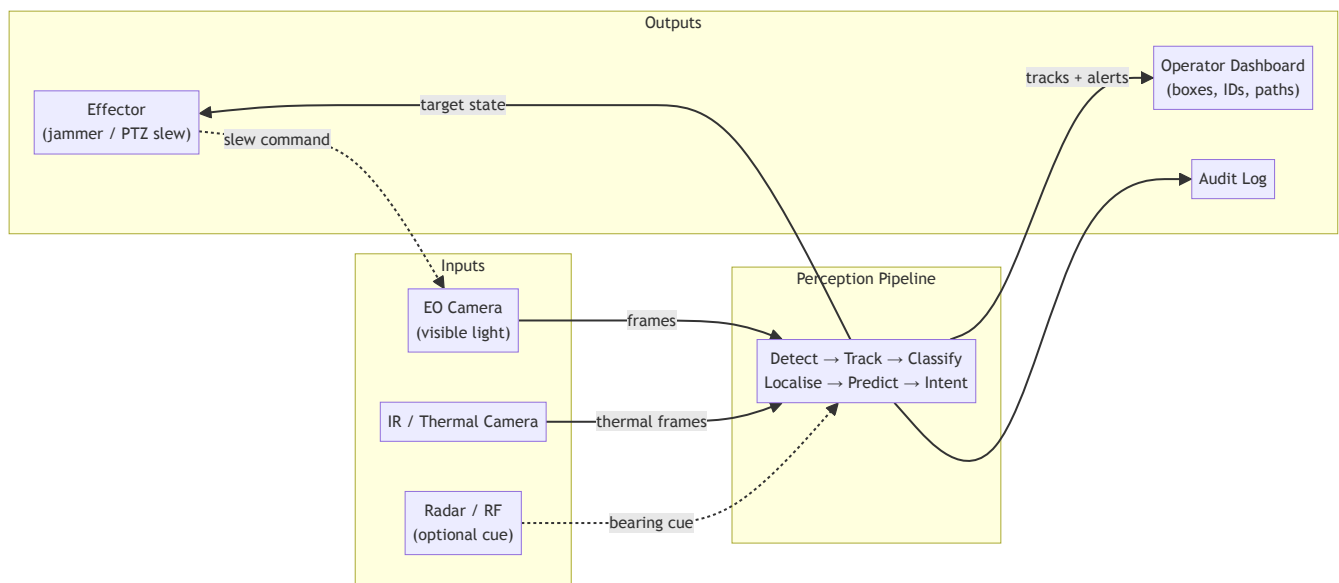
Overview

A real-time pipeline that detects drones in a camera feed, tracks each one across frames, and forecasts its path 1 to 2 seconds ahead. It runs at about 48 fps on a laptop.

- **Detect:** find drone-sized objects in every frame
- **Track:** hold a stable identity per drone
- **Predict:** forecast where each drone is heading

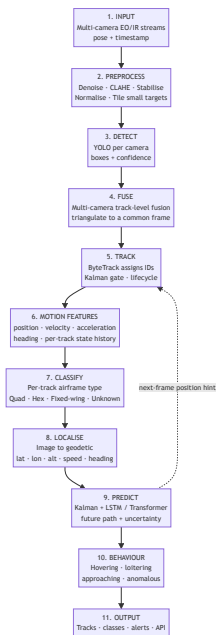
Scope. The diagrams show the full fielded design (multi-camera EO/IR, edge hardware, fusion). The prototype implements the detect-track-predict core on a single video. Solid arrows are data flow; dashed arrows are feedback.

Diagram 1: System Context



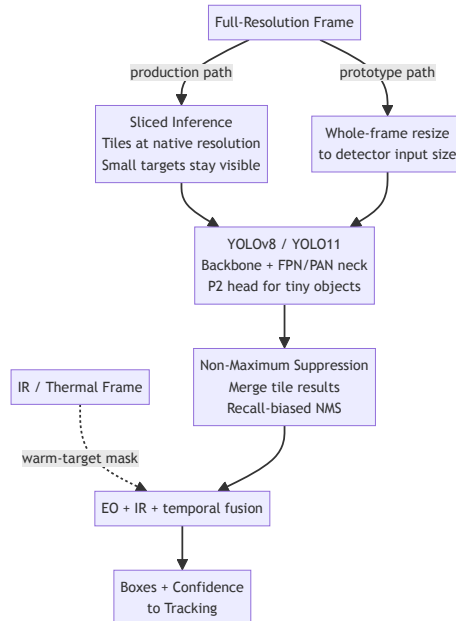
Sensors feed the pipeline; it outputs tracks, alerts, and cueing. The prototype uses one EO video; the rest is the production system around it.

Diagram 2: End-to-End Pipeline



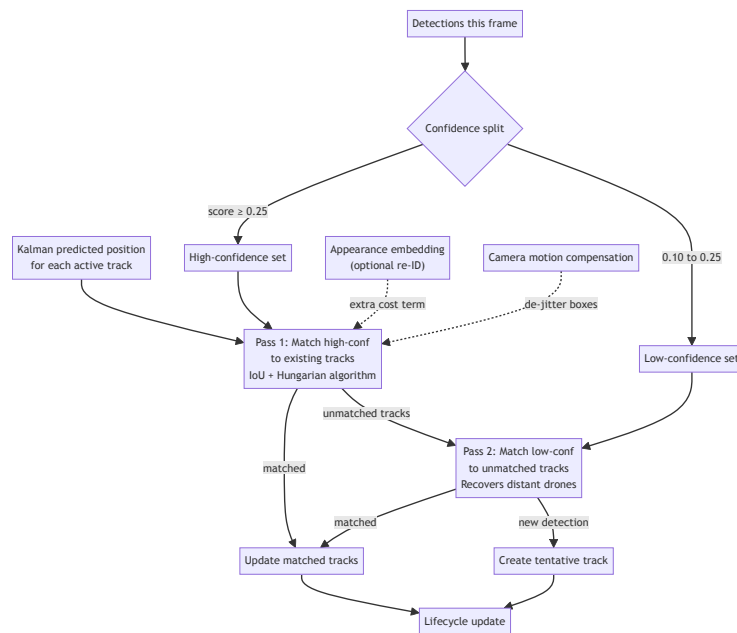
The eleven stages from input to output. The dashed arrow feeds the predictor's next-frame estimate back to the tracker. The prototype implements detection (3), tracking (5), motion-feature extraction (6) and prediction (9) in a single-camera, image-coordinate form; fusion (4), classification (7), localisation (8) and behaviour (10) are production stages, detailed under Production-Grade Extensions.

Diagram 3: Detection Stage



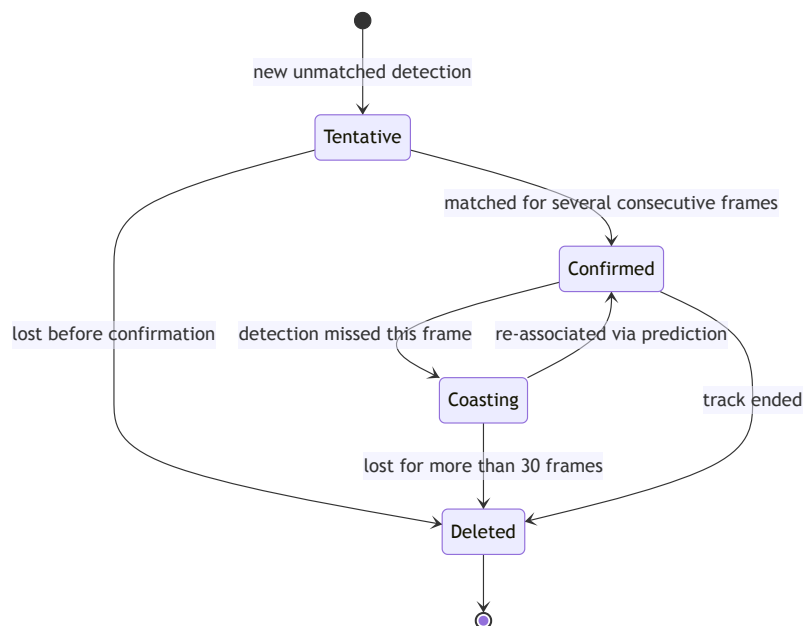
Sliced inference and a P2 head keep small targets detectable. NMS is biased toward recall, because a detection dropped here cannot be recovered downstream. The prototype runs whole-frame YOLO.

Diagram 4: Tracking Stage (ByteTrack)



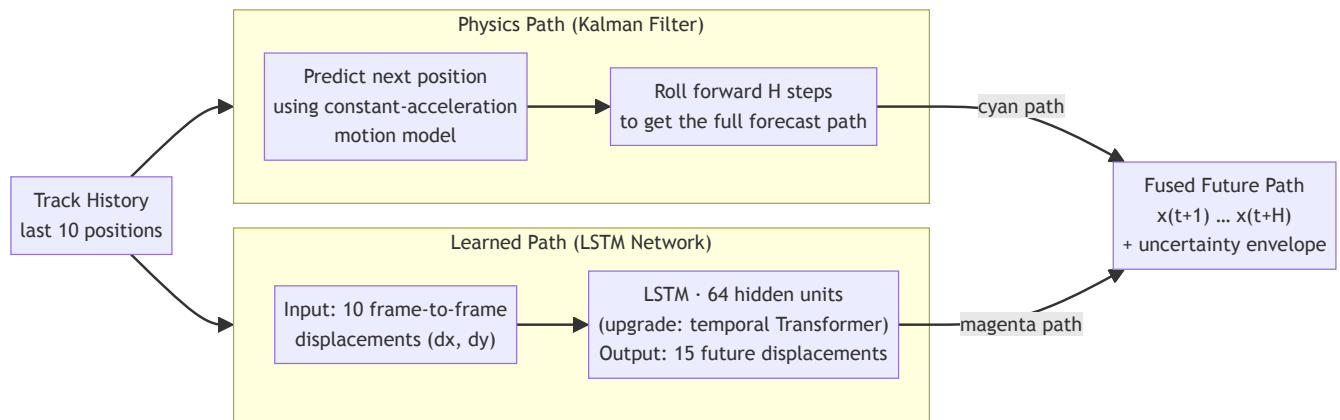
Pass 2 matches the low-confidence boxes other trackers discard, which is where faint, distant drones appear.

Diagram 5: Track Lifecycle



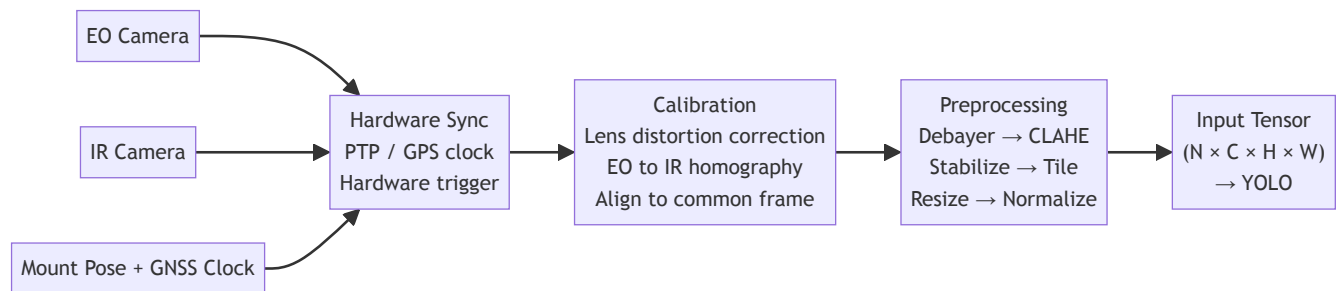
A track must persist several frames to be Confirmed, which filters one-frame false alarms. Coasting holds a track's identity for up to 30 frames through brief occlusion.

Diagram 6: Prediction Stage



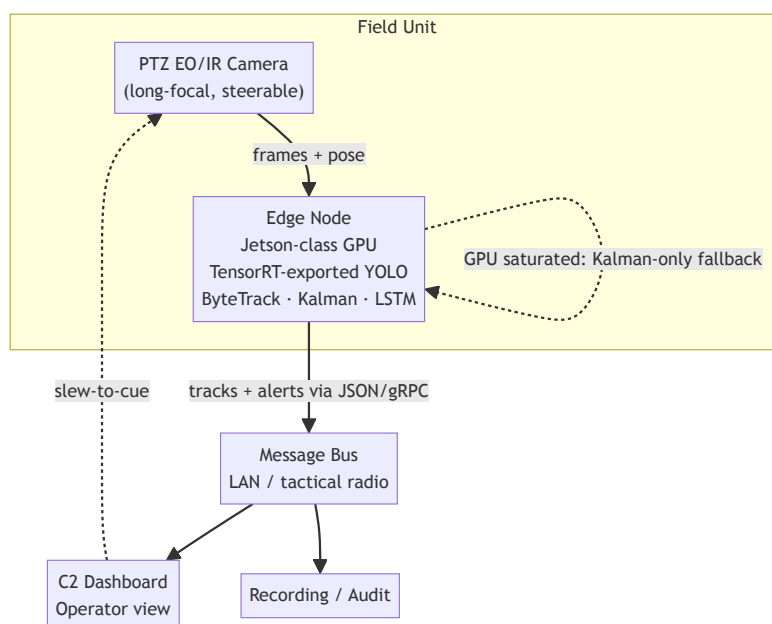
Two forecasts run in parallel: the Kalman filter (physics, stable) and the LSTM (learned, follows manoeuvres). Production upgrade: a temporal Transformer for swarms and longer horizons, with the filter as the deterministic anchor.

Diagram 7: Input Pipeline and Sensor Synchronisation



Hardware-synchronised EO/IR, calibrated and registered, then preprocessed into the input tensor. The prototype runs resize and normalise only.

Diagram 8: Deployment Architecture



An edge node (Jetson-class, TensorRT) feeds a message bus to command-and-control. Degraded mode: if the GPU saturates, the Kalman filter runs alone rather than stalling.

Latency Budget

End-to-end per-frame timing (Apple M3, YOLOv8n at 640 px, 432-frame average):

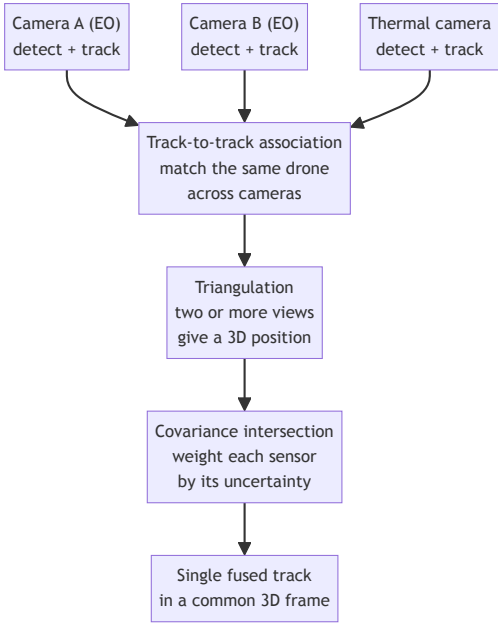
Stage	Time
Capture + preprocess	~1.0 ms
YOLO detect	~6.4 ms
ByteTrack + Kalman	~1.0 ms
LSTM predict	~1.0 ms
Draw + encode	~11.0 ms
End-to-end	~20.9 ms (47.9 FPS)

The 30 fps budget is 33 ms, leaving about 12 ms of headroom for EO/IR fusion and tiling in production.

Production-Grade Extensions

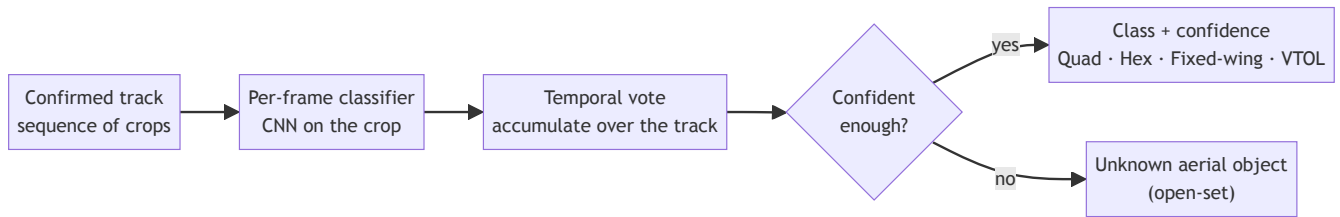
The stages below take the core pipeline from a working demo toward a fielded system. They are part of the architecture, not the prototype code.

Diagram 10: Multi-Camera Fusion



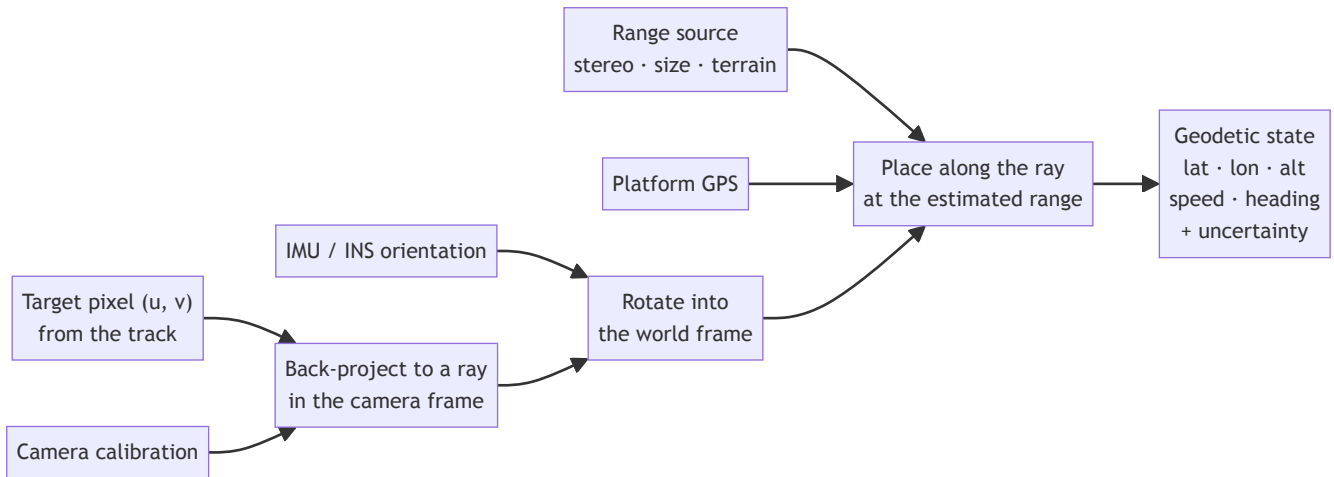
Late, track-level fusion across separated cameras. Triangulation recovers 3D position; covariance intersection weights each sensor by its uncertainty.

Diagram 11: Target Classification



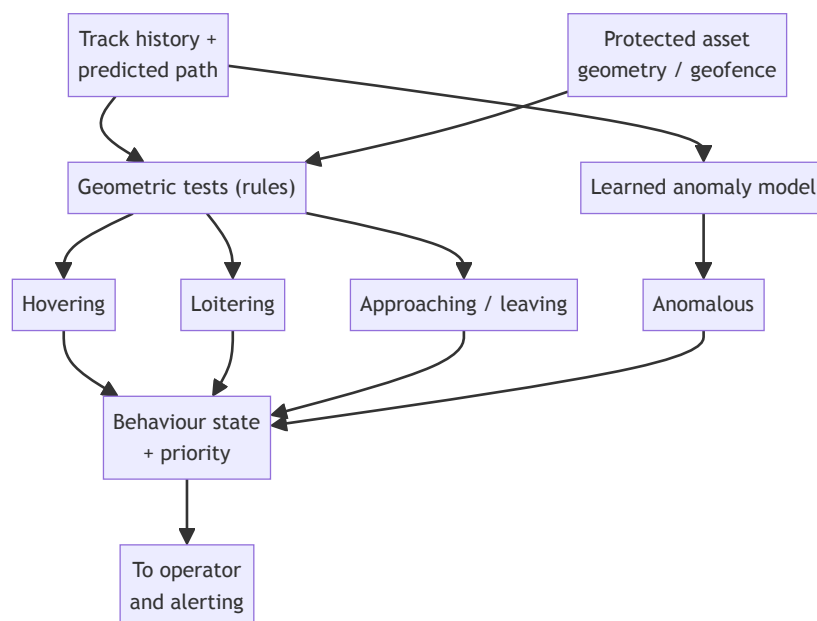
Classification runs per track, not per frame: a temporal vote builds confidence. An open-set unknown class flags unfamiliar airframes instead of mislabelling them.

Diagram 12: World-Coordinate Estimation



Pixel to geodetic state using calibration, GPS, INS and a range source. Range is the hard part for one camera; the output carries an uncertainty ellipse. This is the largest gap between prototype and deployment.

Diagram 13: Behaviour and Intent Analysis



Hovering, loitering and approach are deterministic geometric rules against the protected asset; anomalous behaviour uses a learned model of normal traffic.

Summary Table

Diagram	Shows	Prototype
1: System context	Sensors in, outputs out	Pipeline only
2: End-to-end pipeline	All 10 stages	Stages 3, 5, 8
3: Detection	YOLO, small-target strategy, NMS	Whole-frame YOLO
4: Tracking	ByteTrack two-pass association	Implemented
5: Track lifecycle	Tentative, Confirmed, Coasting	Implemented
6: Prediction	Kalman + LSTM (upgrade: Transformer)	Implemented
7: Input pipeline	Sensor sync, calibration, preprocess	Resize/normalise
8: Deployment	Edge node, bus, degraded mode	Design only
10: Multi-camera fusion	Track-level fusion, triangulation	Design only
11: Classification	Per-track airframe type, unknown class	Design only
12: World coordinates	Image to geodetic, with uncertainty	Design only
13: Behaviour / intent	Hover, loiter, approach, anomaly	Design only